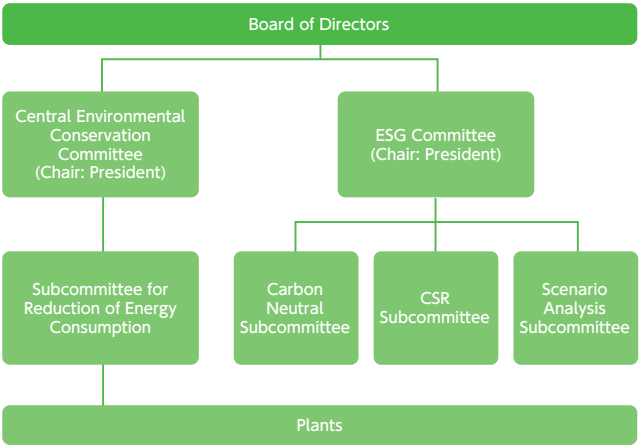




Environmental Management of the NOK Group

With the Paris Agreement adopted in 2015 and the SDGs agreement towards 2030, society's perception of environmental issues is changing. As activities to realize a sustainable society become more active, companies are strongly urged to carry out "environmental management," which is a business activity that takes the environment into account. This includes "companies' disclosure of climate-related financial disclosure" recommended by the Task Force on Climate-related Financial Disclosures (TCFD) and the Science Based Targets (SBTs), which is scientific CO<sub>2</sub> emission reduction targets in conformity with the realization of the Paris Agreement. The NOK Group established the NOK Twin Green Plan 2030, a long-term environmental vision for the year 2030 as its environmental management, and has been engaged in environmental conservation activities such as measures against climate change, resource conservation and recycling, conservation of water resources, measures on substances of concern, development of environmentally friendly products, and conservation of biodiversity. To realize a sustainable society, we will continue to fulfill our social responsibility by promoting environmentally friendly business activities and provide environmental value to society by providing products that contribute to the environment.

Formulation of Long-term Environmental Vision for 2030

Amid the worsening issues of climate change and energy caused by global warming, the NOK Group formulated the "NOK Twin Green Plan 2030" in 2018 in response to the SDGs adopted by the United Nations, and has since been promoting environmental conservation activities. However, with the emergence of global climate change,

which further requires measures against climate change, the NOK Group has decided to review its total CO<sub>2</sub> emissions targets and promote environmental conservation activities in light of international environmental conditions such as the Paris Agreement, TCFD, and SBT.

NOK Twin Green Plan 2030

Green Factory

Through measures for environmentally friendly production, we will reduce CO<sub>2</sub> emissions from factories by 30% from the FY 2018 level.

Initiatives

- We will develop production facilities that will reduce energy consumption by 20% compared with the existing ones.
- We will actively promote the use of renewable energy.
- We will strive for waste reduction and a 100% recycling rate.
- We will work on reducing water use and ensure cleaner wastewater.
- We will thoroughly oversee and reduce the use of hazardous chemicals in processes.

Relevant SDGs



Green Product

We will promote the development of products that will contribute to next-generation environmentally friendly technologies.

Initiatives

- We will develop products that contribute to the spread of next generation energy-saving technologies such as fuel cells.
- We will contribute to the reduction of environmental impact by developing low friction technologies that will make low fuel consumption possible.
- We will develop functional membranes that will contribute to water recycling (hygienic water supply and circulation).
- We will thoroughly control chemical substances in our products and improve their environmental quality.

Relevant SDGs



NOK's Mid-Term Goals and Progress

UNGC  
Principle8

The NOK Group established the Eighth Corporate Environmental Conservation Targets (FY 2020 to FY 2022) for the following categories: environmental management, measures against climate change, resource conservation and recycling, water resource conservation, measures against environmentally hazardous substances, and development of environmentally friendly products. In FY 2020, we have established a department for environmental risk assessment and environmental improvement. The department has conducted and completed environmental risk surveys

at all our plants. Our CO<sub>2</sub> emissions and electricity consumption are being steadily reduced to reach our targets. However, due to the significant impact of a decrease in our production due to the novel coronavirus pandemic, we were unable to achieve our target for CO<sub>2</sub> emissions intensity. On the other hand, we are able to meet the target levels for waste recycling rate, control of increase in water withdrawals, and reduction of VOC emissions. For details, see the values of the results.

The Eighth Corporate Environmental Conservation Targets (FY 2020 to FY 2022)

Slogan: Creating Corporate Value by Increasing Environmental Value

| Category                                  | Initiatives and Goals                                                                                                                                                                                               |                   |                      |                                                                         | Evaluation | Result |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------------|-------------------------------------------------------------------------|------------|--------|
|                                           | Item                                                                                                                                                                                                                | Scope of Coverage | Baseline Fiscal Year | Target                                                                  |            |        |
| Environmental Management                  | <b>Fostering and establishing an environmental climate</b><br>Promoting environmental contribution activities tailored to the characteristics of each region                                                        |                   |                      |                                                                         | ○          | P.58   |
|                                           | <b>Improving the management level of the environmental management system</b><br>Promoting and expanding education by rank and developing environmental human resources                                              |                   |                      |                                                                         |            |        |
|                                           | Reducing environmental protection risks through a steady PDCA cycle                                                                                                                                                 |                   |                      |                                                                         |            |        |
|                                           | <b>Strengthening the prevention of environmental nonconformity</b><br>Promoting the environmental risk assessment of equipment and facilities                                                                       |                   |                      |                                                                         |            |        |
| Measures against Climate Change           | <b>Reducing CO<sub>2</sub> emissions</b><br>Developing energy-saving equipment for production processes<br>Reducing CO <sub>2</sub> emissions in logistics processes<br>Actively introducing renewable energy       |                   |                      |                                                                         |            | P.60   |
|                                           | Electricity Consumption                                                                                                                                                                                             | Japan             | FY 2018              | Reduced by 10%                                                          | ○          |        |
|                                           | CO <sub>2</sub> emissions                                                                                                                                                                                           | Global            | FY 2018              | Reduced by 10%                                                          | ○          |        |
|                                           | CO <sub>2</sub> emissions intensity                                                                                                                                                                                 | Japan             | Previous year        | Average intensity change reduced by 1% or more over the past five years | ×          |        |
| Resource Conservation and Waste Recycling | <b>Resource conservation</b><br>Promoting resource conservation through design and process changes, optimization of material input, etc.                                                                            |                   |                      |                                                                         |            | P.63   |
|                                           | <b>Waste reduction</b><br>Promoting product non-conformity measures and recycling, reducing the amount of waste disposal in landfills by converting waste to recyclable materials, and improving the recycling rate |                   |                      |                                                                         |            |        |
|                                           | Waste Recycling Rate                                                                                                                                                                                                | Japan             | -                    | Maintaining at rate of more than 98%                                    | ○          |        |
| Conservation of Water Resources           | Checking the volume of water withdrawals and wastewater in and outside Japan<br>Controlling the increase in total water withdrawals                                                                                 |                   |                      |                                                                         |            | P.65   |
|                                           | Total water withdrawals                                                                                                                                                                                             | Global            | Previous year        | Controlling an increase                                                 | ○          |        |
| Measures on Substances of Concern         | <b>Reducing environmental burden</b><br>Reducing VOC emissions                                                                                                                                                      |                   |                      |                                                                         |            | P.67   |
|                                           | <b>Strengthening the chemical substance management system</b><br>Promoting green procurement (thorough management of chemical substances by suppliers)                                                              |                   |                      |                                                                         |            |        |
|                                           | VOC emissions                                                                                                                                                                                                       | Global            | FY 2010              | Less than FY 2010 results                                               | ○          |        |
| Environmentally friendly products         | Developing environmentally friendly products<br>Improving the level of environmental conservation assessment in products                                                                                            |                   |                      |                                                                         | ○          | P.68   |

Material Flow

Input materials and industrial waste associated with business activities

| INPUT                                                                                    |        |                      | FY 2020 NOK Group                                                                                                                                                     |               | OUTPUT                                                                                                          |                      |                            |
|------------------------------------------------------------------------------------------|--------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------|----------------------|----------------------------|
| Raw Materials                                                                            |        |                      | Design                                                                               |               | Gas emissions                                                                                                   |                      |                            |
| Total raw material input                                                                 | 81     | 1,000 tons           | •Environmental Conservation Assessment of Products, etc.<br>•Environmental Impact Assessment                                                                          |               | Total CO <sub>2</sub> emissions (Scope 1 + Scope 2)* <sup>1</sup>                                               | 640                  | 1,000 tons-CO <sub>2</sub> |
| Rubber                                                                                   | 12     | 1,000 tons           |                                                                                                                                                                       |               | Scope 1 (Fuel)                                                                                                  | 38                   | 1,000 tons-CO <sub>2</sub> |
| Additives & Adhesives                                                                    | 10     | 1,000 tons           |                                                                                                                                                                       |               | Scope 2 (Electricity/heat)                                                                                      | 602                  | 1,000 tons-CO <sub>2</sub> |
| Plastics                                                                                 | 1      | 1,000 tons           |                                                                                                                                                                       |               | Scope 3 (Indirect Emissions)                                                                                    | 268,547              | 1,000 tons-CO <sub>2</sub> |
| Metals                                                                                   | 58     | 1,000 tons           |                                                                                                                                                                       |               |                                                                                                                 |                      |                            |
| Energy                                                                                   |        |                      | Procurement                                                                          |               | NO <sub>x</sub> (Nitrogen Oxides) * <sup>2</sup>                                                                | 23                   | t                          |
| Total energy input                                                                       | 11,899 | TJ                   | •Environmental Impact Assessment<br>•Green Procurement<br>•Environmental Conservation Inspection of Machinery and Equipment                                           |               | SO <sub>x</sub> (Sulfur Oxides) * <sup>2</sup>                                                                  | 3                    | t                          |
| Purchased power                                                                          | 11,273 | TJ                   |                                                                                                                                                                       |               | VOC (Volatile Organic Compounds)                                                                                | 3                    | 1,000 tons                 |
| Gasoline                                                                                 | 12     | TJ                   |                                                                                                                                                                       |               | Wastewater                                                                                                      |                      |                            |
| Kerosene                                                                                 | 7      | TJ                   |                                                                                                                                                                       |               | Total wastewater                                                                                                | 6,484                | 1,000 m <sup>3</sup>       |
| Heavy Oil A                                                                              | 137    | TJ                   |                                                                                                                                                                       |               | River / Lake                                                                                                    | 2,602                | 1,000 m <sup>3</sup>       |
| Light Oil                                                                                | 150    | TJ                   |                                                                                                                                                                       | sawage system | 3,881                                                                                                           | 1,000 m <sup>3</sup> |                            |
| Liquefied Petroleum Gas (LPG)                                                            | 109    | TJ                   | Production                                                                         |               | BOD load * <sup>3</sup>                                                                                         | 5.4                  | t                          |
| City Gas                                                                                 | 212    | TJ                   | •Life Cycle Assessment (LCA)<br>•Pollution Prevention<br>•Chemical Substances Control<br>•Energy Conservation<br>•Waste Reduction<br>•Environmental Impact Assessment |               | Industrial waste, etc.                                                                                          |                      |                            |
| Water                                                                                    |        |                      |                                                                                                                                                                       |               | By-product generation                                                                                           | 83                   | 1,000 tons                 |
| Total water withdrawals                                                                  | 8,254  | 1,000 m <sup>3</sup> |                                                                                                                                                                       |               | Industrial Waste                                                                                                | 46                   | 1,000 tons                 |
| City Water                                                                               | 4,488  | 1,000 m <sup>3</sup> |                                                                                                                                                                       |               | Amount of waste recycled                                                                                        | 41                   | 1,000 tons                 |
| Industrial Water                                                                         | 3,509  | 1,000 m <sup>3</sup> |                                                                                                                                                                       |               | Final disposal                                                                                                  | 5                    | 1,000 tons                 |
| Ground Water                                                                             | 257    | 1,000 m <sup>3</sup> | Logistics                                                                          |               | Valuable waste                                                                                                  | 37                   | 1,000 tons                 |
| Environmentally hazardous substances                                                     |        |                      | •Efficient Transportation<br>•Promotion of Modal Shift                                                                                                                |               | Environmentally hazardous substances                                                                            |                      |                            |
| Chemical substances subject to the Pollutant Release and Transfer Register (PRTR) system | 5.6    | 1,000 tons           |                                                                                                                                                                       |               | Chemical substances subject to the Pollutant Release and Transfer Register (PRTR) system (discharge volume)     | 127                  | tons                       |
|                                                                                          |        |                      |                                                                                                                                                                       |               | Chemical substances subjected to the Pollutant Release and Transfer Register (PRTR) system (amount of movement) | 323                  | tons                       |
|                                                                                          |        |                      | Customers                                                                          |               |                                                                                                                 |                      |                            |

\* 1 Total CO<sub>2</sub> emissions: Calculated based on the "Greenhouse gas emissions accounting and reporting manual." The emission factor is based on the national emission factor by country published by the International Energy Agency.

\* 2 NO<sub>x</sub> (nitrogen oxides) and SO<sub>x</sub> (sulfur oxides): Total amount of NO<sub>x</sub> and SO<sub>x</sub> generated from specified facilities under the Air Pollution Control Act.

\* 3 BOD load: Biochemical oxygen demand (BOD) is one of the indicators of water pollution and represents the amount of oxygen used by microorganisms to decompose pollutant organic material in water. The figure for BOD load was obtained by multiplying the BOD measurement by the amount of effluent to rivers.

Environmental Management System

UNGC  
Principle7

NOK establishes a system in accordance with the ISO 14001 standard, promotes environmental management, and strives to implement environmental management with the next generation in mind, thereby reducing environmental burden. The current percentage of the NOK Group establishments with ISO 14001 certification is 98.5%.

Environmental audit

At NOK, the Environmental Management Office is in charge of environmental audit and periodically checks and evaluates the operation of the management system, compliance with environmental laws and regulations, and the management of environmentally hazardous substances. The results of the audit are reported to the president and heads of business units for prompt correction and improvement. The company strives to improve its environmental performance and reduce environmental risks by reflecting the results in our environmental protection management policy as necessary.



Environmental audit by the Environmental Management Office staff

External Inspection

NOK has a third-party organization inspect mainly the company's production sites involving in environmental burden and have them acquire and maintain the ISO 14001 certification. In fiscal 2020, there were no significant findings and the company received an evaluation that the management system is operating properly and undergoing continuous improvement. In addition, the president attends the annual meeting where inspectors report on the results of their inspections, so that the valuable opinions of the third party on NOK's environmental management can be reflected in management.

Environmental Education

The NOK Group conducts a variety of education programs to foster a company-wide environmental climate. The NOK Group conducts education by rank for its employees and specialized education including ISO 14001 management, energy, waste, and chemical substance management, as well as education in preparation for emergency situations, thereby, raise the environmental awareness of every employee.

FY 2020 results

| Item                     |                                       | Contents                                                                                      | Number of attendees |
|--------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------|---------------------|
| Level-specific education | New employee training                 | Basic education on environmental conditions and corporate environmental protection activities | 97                  |
|                          | General employee education            | Environmental conditions and the status of NOK, environmental protection activities, etc.     | 7,038               |
|                          | Education for management position     | Environmental conditions and the status of NOK, environmental protection activities, etc.     | 553                 |
| Specialized education    | Emergency-related education           | Education for environmental accidents and emergency response                                  | 1,973               |
|                          | Education for internal auditors       | Education for ISO 14001 auditors                                                              | 4,286               |
|                          | Waste-related education               | Proper waste disposal                                                                         | 5,991               |
|                          | Energy-related course                 | Course for energy managers                                                                    | 120                 |
|                          | Chlorofluorocarbon-related course     | Act on Rational Use and Appropriate Management of Fluorocarbons etc.                          | 15                  |
|                          | Chemical substances-related education | Education on the management of toxic and hazardous materials                                  | 3,273               |



Environmental Communication

Environmental Award System

In FY 2020, the NOK Group established the Environmental Award System, aiming to revitalize internal environmental activities and raise the environmental awareness of each and every employee. Through the Environmental Award System, we are sharing information on outstanding environmental efforts to raise our environmental awareness.

In FY 2020, Thai NOK Co., Ltd., which promoted energy conservation, CO<sub>2</sub> reduction, efficient use of water resources, and tree-planting activities under "Total ESG," won the Gold Award. Their environmental activities and community contribution activities, such as cashing in valuable resources obtained from picking up trash and donating the money to local schools and hospitals, were admired.

Through the Environmental Award System, we will continue to contribute to the realization of a sustainable society by raising our environmental awareness and promoting better environmental activities.



The staff of Thai NOK, the Gold Award winner

Environmental Improvement in Collaboration among Group Companies

In addition to the environmental improvement activities our respective bases carried out, NOK established a dedicated department for environmental risk assessment and improvement in 2020. The staff members of the department visit our 48 bases in Japan and carry out risk assessments and environmental improvement activities together with the staff members of the base to promote energy conservation and prevent environmental accidents. By conducting risk assessments and improvement activities across boundaries between bases, we are striving to share environmental information, encourage other bases to adopt efficient improvement activities, and bring up the levels of the environmental improvement activities at the NOK Group. NOK will continue to contribute to reducing the environmental burden caused by the group through our group-wide environmental improvement activities.

External Awards / Commendations

The Fukushima Plant received the highest award in the manufacturing industry category of the Fukushima Protocol compliant project commendation for their daily environmental activities to reduce CO<sub>2</sub> emissions, waste, and water usage.

MYK Corporation has been evaluated as Rank S for the fifth consecutive year under the business classification system of the Ministry of Economy, Trade and Industry. In addition, we received the Chubu Electricity Use Rationalization Committee Chairman's Award as an excellent energy management company for developing an injection molding machine with high energy-saving performance through use of a subsidy.

We will continue to actively develop environmental activities to contribute to local communities and the environment and increase our corporate value.



Commendation ceremony for Fukushima Protocol compliant projects



MYK Corporation

Status of compliance with environmental laws and regulations

During FY 2020, the NOK Group was involved in no fines, penalties or lawsuits related to environmental incidents or complaints.

Efforts to reduce the environmental load in our supply chain

Since FY 2020, NOK has been conducting a survey with our suppliers regarding their environmental efforts in order to work together in the supply chain to reduce the environmental load. Through the survey, we confirm the status of their efforts to deal with climate change and water risk, reduction of CO<sub>2</sub> emissions, and water risk assessment. Also, we are working to raise environmental awareness in our supply chain with supplementary materials on such topics as environmental trends and risk assessment methods. We will continue to promote the survey to raise environmental awareness in our supply chain, and cooperate in promoting activities for environmental load reduction.

Measures against Climate Change

Social Background

In recent years, there has been concern that typhoons, heavy rains, droughts, heat waves, and other abnormal weather will occur more frequently around the world, threatening the survival of humanity in the future. In 2016, the Paris Agreement came into force at the United Nations, and efforts to reduce greenhouse gases such as CO<sub>2</sub> and energy conservation have been gradually increasing. The number of countries proclaiming that they are aiming to achieve carbon neutrality by 2050 has started to increase, and the world is getting ready to move toward carbon neutrality together. In Japan, the government declared "Carbon Neutral 2050" in October 2020, and other countries have started promoting their efforts for decarbonization. Efforts for decarbonization are becoming important for companies as a factor that influences their operations. It is necessary for companies to conduct active efforts for the reduction of greenhouse gases, such as energy saving and the introduction of renewable energy based on long-term strategies, and disclosure of information about the activities.

NOK Group Policy

The NOK Group will work to reduce emissions of CO<sub>2</sub>, which is a major greenhouse gas, for the prevention of global warming. Our manufacturing departments have developed energy-saving equipment to reduce production-related energy, and the logistics department has been reducing fuel consumption for transportation equipment. We take measures to reduce CO<sub>2</sub> emissions across the lifecycle of our products, including the introduction of renewable energy throughout the company. In addition, we regard climate change as an important issue in our business management, and the NOK Group is planning a management strategy against climate change in line with the TCFD recommendations.

Targets and Results

In FY 2020, the CO<sub>2</sub> emissions of the NOK Group were reduced by 17.2% from the FY 2018 level, and it meets the target of a "30%

reduction in CO<sub>2</sub> emissions from the FY 2018 level" stated in the NOK Twin Green Plan 2030. The reduction achieved this year is, however, considered to be caused by the decrease in production due to the novel coronavirus pandemic. For the CO<sub>2</sub> emissions intensity, NOK was unable to meet its target, or the average intensity change reduced by 1% or more over the past five years, which resulted in reduction by 0.12%. NOK keeps striving to reduce CO<sub>2</sub> emissions on a global basis and aims to reduce our CO<sub>2</sub> emissions intensity through improvement in production efficiency and other measures.

General

- Switching to LED lighting
- Frequently turning off lights, automatic light activation

Logistics

- Promotion of idle stop
- Optimization of distribution routes
- Improvement of vehicle fuel efficiency
- Reducing fuel consumption by improving loading rates

Production

Factory building

- Upgrading to high efficiency equipment
- Optimization of air conditioning operation
- Introduction of renewable energy

Production facilities

- Improvement of production efficiency
- Retrofit
- Power saving of facilities
- Control and reuse of heat dissipation and waste heat
- Shifting motors and pumps, etc. to inverter type

Motor

- Repair of air leak
- Optimization of operating conditions
- Upgrading to high efficiency equipment

Process monitoring

- Visualization of power consumption
- Visualization of operational status

| Target                                                       | Baseline Fiscal Year | Baseline Figure                            | Target Fiscal Year | Target Figure                                                                                                                                  | This fiscal year's results                                                                                                                        | Evaluation                                                                                                                |
|--------------------------------------------------------------|----------------------|--------------------------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| CO <sub>2</sub> emissions Reduction Target (Global)          | 2018                 | 774 [1,000 tons-CO <sub>2</sub> ]          | 2030               | Reduction by 30% CO <sub>2</sub> emissions 542 [1,000 tons-CO <sub>2</sub> ] or below                                                          | Reduced by 17.2% CO <sub>2</sub> emissions 640 [1,000 tons-CO <sub>2</sub> ]                                                                      | ○ CO <sub>2</sub> emissions Achieved 2.5% or more/year                                                                    |
| CO <sub>2</sub> emissions Intensity Reduction Target (Japan) | 2019                 | 1,330 [ton-CO <sub>2</sub> /1 million yen] | 2022               | Average change reduced by 1% or more over the past five years (CO <sub>2</sub> emissions intensity 1.161 [ton-CO <sub>2</sub> /1 million yen]) | Average change reduced by 0.12% or more over the past five years (CO <sub>2</sub> emissions intensity 1.220 [ton-CO <sub>2</sub> /1 million yen]) | × CO <sub>2</sub> emissions intensity Average intensity change reduced over the past five years 1% reduction not achieved |

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Concrete Activities

Measures for Energy Conservation (Japan)

The NOK Group is replacing fluorescent lamps with LED lamps, repairing air leaks, and taking other minor energy-saving measures. Kyushu Plant of NOK Fugaku Engineering, one of our group companies, replaced fluorescent lamps in the manufacturing buildings with LED lamps, and successfully reduced the amount of electricity consumption by 1,493 kWh/month. Also, we have repaired all air leaks in all plants, started power cutoffs on weekday nights and holidays at the headquarters plant, and stopped the unnecessary operation of air conditioners. These efforts reduced the operating load of the compressors, causing the amount of electricity consumption to be reduced by 132 thousand kWh (a 25.1% reduction from the previous year) and the energy used for air conditioners to be reduced by 30.3% from the previous year.

Improvement for Air Conditioning (China)

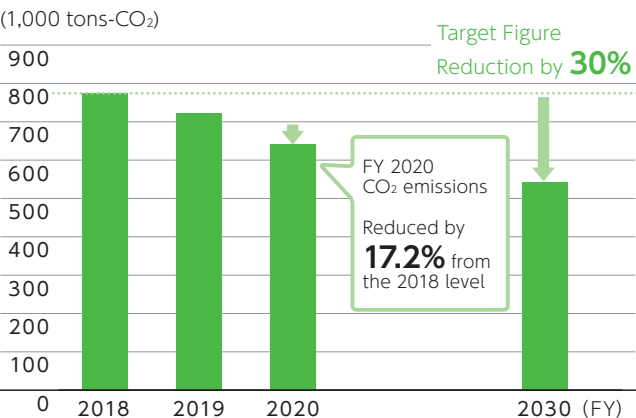
Wuxi NOK-Freudenberg Oilseal Co., Ltd. set air conditioner air outlets 1.8 meters closer to the floor at their production site to save energy on air conditioning. Through the lowered air outlets, cold air can get to the workers without loss, and the on-site temperature during summer was successfully lowered by 3°C. The improvement resulted in annual energy savings of 15,120 kWh and the reduction of CO<sub>2</sub> emissions by 141.52 t-CO<sub>2</sub>.



Utilization of Waste Heat (China)

Mektect Manufacturing Corporation (Zhuhai) Ltd. makes effective use of waste heat generated in their plant. The waste heat generated by the air compressors is recycled in the dormitory's hot water supply system, and it is expected to save 146,760 kWh annually. We will continue to promote effective use of energy through the recovery of waste heat as one of our energy saving measures.

Total CO<sub>2</sub> emissions of the NOK Group

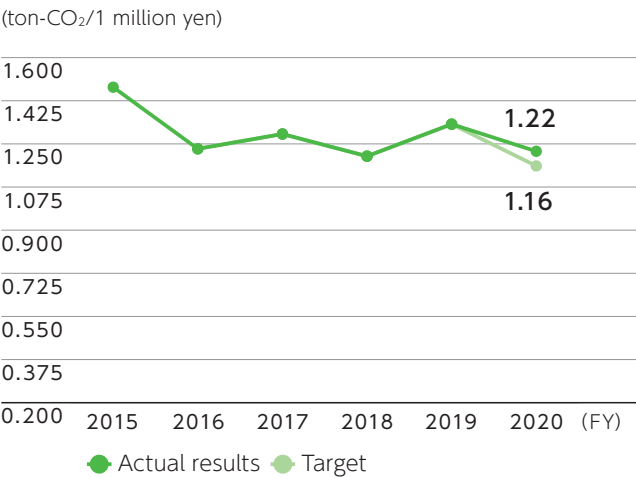


FY 2020 Scope 3 Breakdown

| Category                                                            | CO <sub>2</sub> emissions (1,000 tons-CO <sub>2</sub> ) |
|---------------------------------------------------------------------|---------------------------------------------------------|
| Purchased goods and services 168                                    | 204                                                     |
| Capital goods                                                       | 22                                                      |
| Fuel- and energy-related activities (not included in Scope 1 and 2) | 0                                                       |
| Upstream transportaion and distribution                             | 10                                                      |
| Waste generated from operation                                      | 14                                                      |
| Business travel                                                     | 5                                                       |
| Employee commuting                                                  | 15                                                      |
| Upstream leased assets                                              | 0                                                       |
| Downstream transportation and distribution                          | —                                                       |
| Processing of sold products                                         | —                                                       |
| Use of sold products                                                | —                                                       |
| End-of-life treatment of sold products                              | —                                                       |
| Downstream leased assets                                            | 0                                                       |
| Franchises                                                          | 0                                                       |
| Investments                                                         | —                                                       |
| Total                                                               | 269                                                     |

\* Categories 9, 10, 11, 12, and 15 fall outside the aggregation range because of the current difficulty in calculating.

NOK Group CO<sub>2</sub> Intensity in Japan



Greenhouse Gas Breakdown

| Greenhouse gas                                    | CO <sub>2</sub> emissions (t-CO <sub>2</sub> ) | GWP reference                                                       |
|---------------------------------------------------|------------------------------------------------|---------------------------------------------------------------------|
| CO <sub>2</sub> emissions not from energy sources | 33                                             | IPCC Fourth Assessment Report (AR4 value for 100-year time horizon) |
| CH <sub>4</sub>                                   | 291                                            | IPCC Fourth Assessment Report (AR4 value for 100-year time horizon) |
| N <sub>2</sub> O                                  | 85                                             | IPCC Fourth Assessment Report (AR4 value for 100-year time horizon) |
| HFCs                                              | 481                                            | IPCC Fourth Assessment Report (AR4 value for 100-year time horizon) |
| PFCs                                              | 0                                              | IPCC Fourth Assessment Report (AR4 value for 100-year time horizon) |
| SF <sub>6</sub>                                   | 89                                             | IPCC Fourth Assessment Report (AR4 value for 100-year time horizon) |
| NF <sub>3</sub>                                   | 0                                              | IPCC Fourth Assessment Report (AR4 value for 100-year time horizon) |

Subcommittee for Reduction of Energy Consumption

Activities of Subcommittee for Reduction of Energy Consumption

Based on the NOK Twin Green Plan 2030, NOK engages in continuous and strategic energy conservation activities with the goal of reducing CO<sub>2</sub> emissions from plants by 30% from FY 2018 through environment-friendly manufacturing initiatives. Last year, the Japanese government declared "Carbon Neutral by 2050," and announced a target of a 46% reduction from the 2013 level by 2030 this year. Therefore, we must engage in more aggressive activities. Regarding our electricity usage last year, the ratio of the electricity consumption that fluctuates according to the production volume to the fixed electricity consumption was 60:40. With the tool, we found that most of the former was character-

ized by being used as heat energy in the manufacturing process. Therefore, we have been developing construction methods and equipment for low energy consumption. For fixed power usage, NOK has continued to increase the use of measures already determined effective and introduced them at its plants. Such measures include the replacement and maintenance of LED lighting and top-runner motors and the introduction of renewable energy. We also have been promoting the establishment of a system for more robust energy conservation management by adopting IoT technology to visualize power consumption at out facilities. We will continue improving our activities through the expansion and new development of energy conservation measures.

Development of an Analysis Technique for Vulcanization Equipment

We developed an analysis technique for the heat in the material loading, vulcanization, and removal processes in rubber vulcanization equipment. It is aimed at finding points that lead to energy saving in vulcanization equipment and predicting the effects of possible measures for the points. Figure 1 illustrates a vulcanization mold model. The materials are put into the cavities in the mold and vulcanized. Figure 2 shows an analysis of how the mold cools while the molded product is taken out of the mold after the mold containing the vulcanized materials is removed from the equipment. Figure 3 shows the temperature changes in the mold cavities identified with the analysis technique. One point for saving energy is finding a way to suppress the temperature drop when taking out the product in the heating and cooling cycle in which the material loading, vulcanization, and removal processes are continuously performed. To find an energy saving measure, we analyzed a way to remove the product with a mold heating plate attached. Figure 4 shows the measurements. We found that the measure helps keep the mold temperature in the cycle. We assume that this measure will save energy by requiring less heating by a heater.

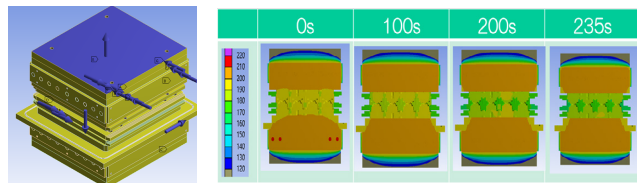


Figure 1. Mold model  
Figure 2. Change of mold temperatures when product is removed

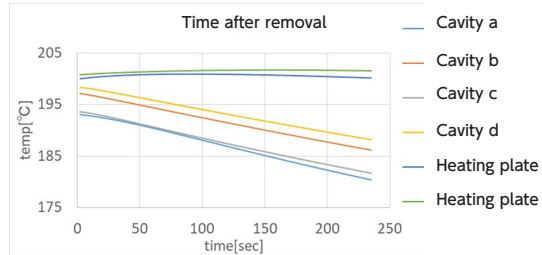


Figure 3. Change of mold cavity temperatures when product is removed

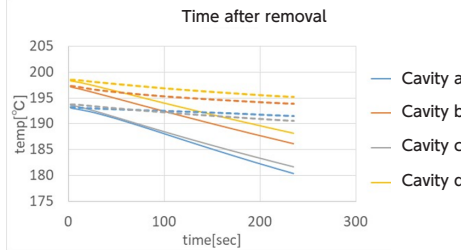


Figure 4. Change of cavity temperatures Solid line: Before improvement, Dashed line: After improvement

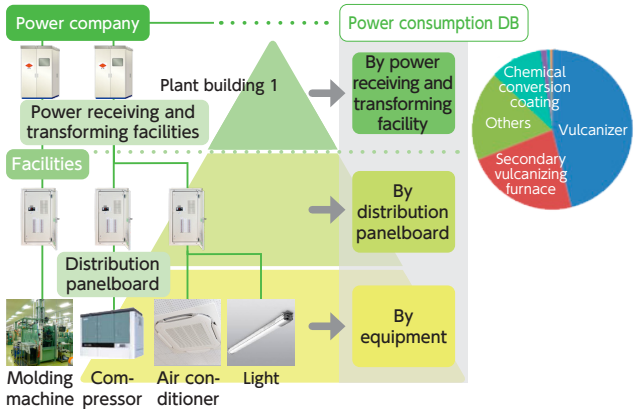


Figure 5. Visualization of power consumption



# Resource Conservation and Waste Recycling

## Social Background

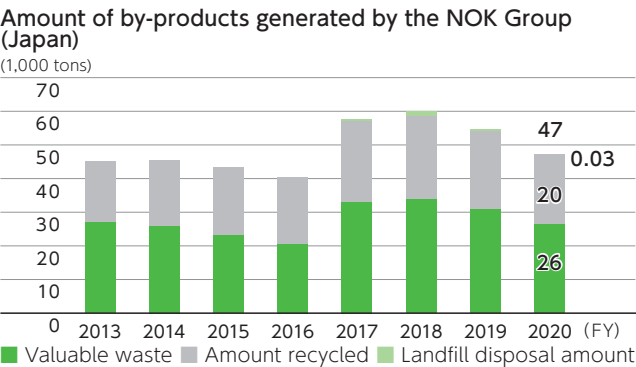
We have been living a convenient and prosperous life surrounded by many things in a social system of mass production, mass consumption and mass disposal. At the same time, we are consuming a large amount of the earth's finite resources and energy, as well as generating a large amount of waste, and we are facing various problems such as the depletion of natural resources and environmental destruction. Today, shifting from a mass-production, mass-consumption, mass-waste society to a recycling society is globally called for, and businesses are required to promote the efficient use of resources taking into account the entire product lifecycle from the procurement of raw materials to disposal and 3Rs (Reuse, Reduce and Recycle).

## NOK Group Policy

The NOK Group has been promoting the 3R initiative (reuse, reduce, and recycle) to realize a recycling-oriented society. The Group strives to contribute to a sustainable society and to enhance our corporate value through a range of measures throughout the product life cycle, including the efficient use of raw materials, water, and energy as well as waste reduction and recycling of resources by improving each process.

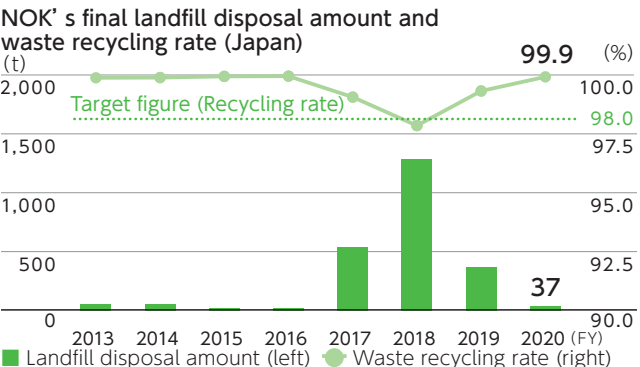
| Target Items                                      | Target Fiscal Year | Target Figure | This fiscal year's results | Evaluation                                      |
|---------------------------------------------------|--------------------|---------------|----------------------------|-------------------------------------------------|
| Continuous efforts to conserve resources          | -                  | -             | -                          | -                                               |
| Industrial waste landfill disposal amount (Japan) | 2030               | 0 t           | 37 t                       | ○<br>Achieved a recycling rate of more than 98% |
| By-product recycling rate (Japan)                 | 2020               | More than 98% | 99.9%                      |                                                 |

\* By-product: General term for industrial waste and valuable resources  
\* Recycling rate: "Recycling rate = 1 - amount of landfill disposal / amount of by-product generated × 100"



## Targets and Results

The NOK Group's recycling rate for FY 2020 was 99.9%, achieving more than 98 percent of its goal. Due to waste (disposed of in landfills) from the foundry in Tottori Prefecture, which started operating in the second half of FY 2017, the amount of waste disposed of in landfill in FY 2018 was 1,241 tons. But in FY 2020, the amount of waste disposed of in landfills was reduced to 37 tons because we were able to recycle it, and the recycling rate was improved to 99.9% of the level recorded before the foundry started operating. The NOK Group will continue to reduce the amount of waste by taking measures to conserve resources, such as reviewing the shape of molds and the appropriate weight of rubber fabric, as well as measures to prevent product non-conformity, promote recycling, and turn waste into valuable resources. Thereby, we will promote the NOK Twin Green Plan 2030 to achieve a 100% recycling rate.



## Concrete Activities

### Reduction of industrial liquid (Japan)

UNIMATEC CO., LTD. has launched a project to move the out-sourced treatment of waste liquid containing fluorine ions (industrial waste) in-house. Through research and development of a system for fluorine ion removal, we found an effective adsorption method and set it with the optimum operating conditions in practical tests to enable in-house treatment. Through our in-house treatment, we have successfully reduced the amount of waste liquid.



### Promotion of recycling (Japan, China)

The NOK Group is actively promoting recycling. At Isshin Industries Corporation, they were able to reduce industrial waste by 240 liters annually by recycling the hydraulic oil used in compression molding machines (recycled hydraulic oil). Mektec Manufacturing Corporation (Suzhou) Ltd. has started purification treatment because copper plating solutions containing impurities are unusable. The treatment allows copper plating solutions to be reused, and 19 tons of copper plating waste solutions were reduced annually.

### Recycling of waste solvents as valuable resources (Japan)

NOK KLUEBER CO., LTD. is recycling waste solvents such as post-cleaning solvents and raw material-containing solvents into valuable resources through cleaning. By distilling and recovering DMF / NMP / IPA / NS Clean (hydrocarbon-based cleaning agent), approximately eight tons of industrial waste were reduced in four months. Also, NIPPON MEKTRON, LTD. has reduced 500 tons of industrial waste by recycling acidic waste liquid as a valuable resource.

### Going paperless (company-wide)

NOK is promoting digitalization company-wide, and one of its goals is to go paperless. We have selected types of paper documents that are printed regularly, such as pay slips, and are aiming to reduce them by 50% by 2022. Of the 220 selected types of printed matter, we have converted 55 types into digital formats. We have successfully reduced the amount of paper consumption by 25%, and the number of paper sheets purchased has been steadily reduced at the respective bases. We will continue to strive to go paperless in order to further reduce waste.

### Reduction in food loss (Fujisawa)

The Shonan R&D Center has begun an activity to reduce food loss in collaboration with an external company operating a cafeteria inside the Center. We investigated the amount and breakdown of daily food waste in the cafeteria, called for adjustment of the amount of rice served, created bulletin boards to raise awareness

of food loss reduction, and checked the change in the amount of waste. The check found that the amount of food waste was reduced by 34% compared to before the activity. We will continue to carry out the activity, and expand and promote the activity so that it is carried out at other bases.

